Choose Your Lenses: Flaws in Gender Bias Evaluation

Hadas Orgad
orgad.hadas@cs.technion.ac.il

Yonatan Belinkov*
belinkov@technion.ac.il

Technion – Israel Institute of Technology

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Abstract

Considerable efforts to measure and mitigate gender bias in recent years have led to the introduction of an abundance of tasks, datasets, and metrics used in this vein. In this position paper, we assess the current paradigm of gender bias evaluation and identify several flaws in it. First, we highlight the importance of extrinsic bias metrics that measure how a model's performance on some task is affected by gender, as opposed to intrinsic evaluations of model representations, which are less strongly connected to specific harms to people interacting with systems. We find that only a few extrinsic metrics are measured in most studies, although more can be measured. Second, we find that datasets and metrics are often coupled, and discuss how their coupling hinders the ability to obtain reliable conclusions, and how one may decouple them. We then investigate how the choice of the dataset and its composition, as well as the choice of the metric, affect bias measurement, finding significant variations across each of them. Finally, we propose several guidelines for more reliable gender bias evaluation.

1 Introduction

A large body of work has been devoted to measurement and mitigation of social biases in natural language processing (NLP), with a particular focus on gender bias (Sun et al., 2019; Blodgett et al., 2020; Garrido-Muñoz et al., 2021; Stanczak and Augenstein, 2021). These considerable efforts have been accompanied by various tasks, datasets, and metrics for evaluation and mitigation of gender bias in NLP models. In this position paper, we critically assess the predominant evaluation paradigm and identify several flaws in it. These flaws hinder progress in the field, since they make it difficult to ascertain whether progress has been actually made.

Gender bias metrics can be divided into two groups: *extrinsic* metrics, such as performance difference across genders, measure gender bias with respect to a specific downstream task, while *intrinsic* metrics, such as WEAT (Caliskan et al., 2017), are based on the internal representations of the language model. We argue that measuring extrinsic metrics is crucial for building confidence in proposed metrics, defining the harms caused by biases found, and justifying the motivation for debiasing a model and using the suggested metrics as a measure of success. However, we find that many studies on gender bias only measure intrinsic metrics. As a result, it is difficult to determine what harm the presumably found bias may be causing. When it comes to gender bias mitigation efforts, improving intrinsic metrics may produce an illusion of greater success than reality, since their correlation to downstream tasks is questionable (Goldfarb-Tarrant et al., 2021; Cao et al., 2022). In the minority of cases where extrinsic metrics are reported, only few metrics are measured, although it is possible and sometimes crucial to measure more.

Additionally, gender bias measures are often applied as a *dataset* coupled with a *measurement technique* (a.k.a metric), but we show that they can be separated. A single gender bias metric can be measured using a wide range of datasets, and a single dataset can be applied to a wide variety of metrics. We then demonstrate how the choice of gender bias metric and the choice of dataset can each affect the resulting measures significantly. As an example, measuring the same metric on the same model with an imbalanced or a balanced dataset[1] may result in very different results. It is thus difficult to compare newly proposed metrics and debiasing methods with previous ones, hindering progress in the field.

To summarize, our contributions are:

- We argue that extrinsic metrics are important

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[1] Balanced with respect to the amount of examples for each gender, per task label.

for defining harms (§2), but researchers do not use them enough even though they can (§5).

- We demonstrate the coupling of datasets with metrics and the feasibility of other combinations (§3).

- On observing that a specific metric can be measured on many possible datasets and vice-versa, we demonstrate how the choice and composition of a dataset (§4), as well as the choice of bias metric to measure (§5), can strongly influence the measured results.

- We provide guidelines for researchers on how to correctly evaluate gender bias (§6).

Bias Statement This paper examines metrics and datasets that are used to measure gender bias, and discusses several pitfalls in the current paradigm. As a result of the observations and proposed guidelines in this work, we hope that future results and conclusions will become clearer and more reliable.

The definition of gender bias in this paper is through the discussed metrics, as each metric reflects a different definition. Some of the examined metrics are measured on concrete downstream tasks (extrinsic metrics), while others are measured on internal model representations (intrinsic metrics). The definitions of intrinsic and extrinsic metrics do not align perfectly with the definitions of allocational and representational harms (Kate Crawford, 2017). In the case of allocational harm, resources or opportunities are unfairly allocated due to bias. Representative harm, on the other hand, is when a certain group is negatively represented or ignored by the system. Extrinsic metrics can be used to quantify both allocational and representational harms, while intrinsic metrics can only quantify representational harms, in some cases.

There are also other important pitfalls that are not discussed in this paper, like the focus on high-resource languages such as English and the binary treatment of gender (Sun et al., 2019; Stanczak and Augenstein, 2021; Dev et al., 2021). Inclusive research of non-binary genders would require a new set of methods, which could benefit from the observations in this work.

2 The Importance of Extrinsic Metrics in Defining Harms

In this paper, we divide metrics for gender bias to three groups:

- **Extrinsic performance:** measures how a model's performance is affected by gender, and is calculated with respect to particular gold labels. For example, the True Positive Rate (TPR) gap between female and male examples.

- **Extrinsic prediction:** measures model's predictions, such as the output probabilities, but the bias is not calculated with respect to some gold labels. Instead, the bias is measured by the effect of gender or stereotypes on model predictions. For example, the probability gap can be measured on a language model queried on two sentences, one pro-stereotypical ("he is an engineer") and another anti-stereotypical ("she is an engineer").

- **Intrinsic:** measures bias in internal model representations, and is not directly related to any downstream task. For example, WEAT.

It is crucial to define how measured bias harms those interacting with the biased systems (Barocas et al., 2017; Kate Crawford, 2017; Blodgett et al., 2020; Bommasani et al., 2021). Extrinsic metrics are important for motivating bias mitigation and for accurately defining "why the system behaviors that are described as 'bias' are harmful, in what ways, and to whom" (Blodgett et al., 2020), since they clearly demonstrate the performance disparity between protected groups.

For example, in a theoretical CV-filtering system, one can measure the TPR gap between female and male candidates. A gap in TPR favoring men means that, given a set of valid candidates, the system picks valid male candidates more often than valid female candidates. The impact of this gap is clear: Qualified women are overlooked because of bias. In contrast, consider an intrinsic metric such as WEAT (Caliskan et al., 2017), which is derived from the proximity (in vector space) of words like "career" or "family" to "male" or "female" names. If one finds that male names relate more to career and female names relate more to family, the consequences are unclear. In fact, Goldfarb-Tarrant et al. (2021) found that WEAT does not correlate with other extrinsic metrics. However, many studies re-

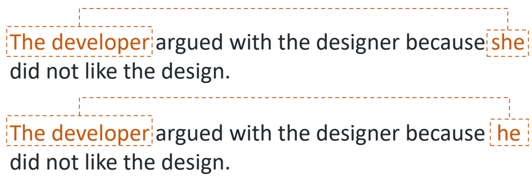

Figure 1: Coreference resolution example from Winobias: a pair of anti-stereotypical (top) and pro-stereotypical examples (bottom). Developers are stereotyped to be males.

port only intrinsic metrics (a third of the papers we reviewed, §5).

3 Coupling of Datasets and Metrics

In this section, we discuss how datasets and metrics for gender bias evaluation are typically coupled, how they may be decoupled, and why this is important. We begin with a representative test case, followed by a discussion of the general phenomenon.

3.1 Case study: Winobias

Coreference resolution aims to find all textual expressions that refer to the same real-world entities. A popular dataset for evaluating gender bias in coreference resolution systems is Winobias (Zhao et al., 2018a). It consists of Winograd schema (Levesque et al., 2012) instances: two sentences that differ only by one or two words, but contain ambiguities that are resolved differently in the two sentences based on world knowledge and reasoning. Winobias sentences consist of an anti- and a pro- stereotypical sentence, as shown in Figure 1. Coreference systems should be able to resolve both sentences correctly, but most perform poorly on the anti-stereotypical ones (Zhao et al., 2018a, 2019; de Vassimon Manela et al., 2021; Orgad et al., 2022).

Winobias was originally proposed as an extrinsic evaluation dataset, with a reported metric of anti- and pro- stereotypical performance disparity. However, other metrics can also be measured, both intrinsic and extrinsic, as shown in several studies (Zhao et al., 2019; Nangia et al., 2020b; Orgad et al., 2022). For example, one can measure how many stereotypical choices the model preferred over anti-stereotypical choices (an extrinsic performance measure), as done on Winogender (Rudinger et al., 2018), a similar dataset. Winobias sentences can also be used to evaluate language models (LMs), by evaluating if an LM gives higher probabilities to pro-stereotypical sentences (Nangia et al., 2020b) (an extrinsic prediction measure). Winobias can also be used for intrinsic metrics, for example as a template for SEAT (May et al., 2019a) and CEAT (Guo and Caliskan, 2021) (contextual extensions of WEAT). Each of these metrics reveals a different facet of gender bias in a model. An explicit measure of how many pro-stereotypical choices were preferred over anti-stereotypical choices has a different meaning than measuring a performance metric gap between two different genders. Additionally, measuring an intrinsic metric on Winobias may be help tie the results to the model's behavior on the same dataset in the downstream coreference resolution task.

3.2 Many possible combinations for datasets and metrics

Winobias is one example out of many. In fact, benchmarks for gender bias evaluation are typically proposed as a package of two components:

1. **A dataset** on which the benchmark task is performed.
2. **A metric**, which is the particular method used to calculate bias of a model on the dataset.

Usually, these benchmarks are considered as a bundle; however, they can often be decoupled, mixed, and matched, as discussed in the Winobias test case above. The work by Delobelle et al. (2021) is an exception, in that they gathered a set of templates from diverse studies and tested them using the same metric.

In Table 1, we present possible combinations of datasets (rows) and metrics (columns) from the gender bias literature. The metrics are partitioned according to the three classes of metrics defined in Section 2. We present only metrics valid for assessing bias in contextualized LMs (rather than static word embeddings), since they are the common practice nowadays. The table does not claim to be exhaustive, but rather illustrates how metrics and datasets can be repurposed in many different ways. The metrics are described in appendix A, but the categories are very general and even a single column like "Gap (Label)" represents a wide variety of metrics that can be measured.

Table 1 shows that many metrics are compatible across many datasets (many ✓'s in the same column), and that datasets can be used to measure a variety of metrics other than those typically mea-

Dataset \ Metric	Extrinsic Performance: Gap (Label)	Gap (Stereo)	Gap (Gender)	Extrinsic Predictions: % or # of Answer Changed	% or # Model Prefers Stereotype	Pred Gap	LM Prediction On Target words	Intrinsic: SEAT	CEAT	Probe	Cluster	Nearest Neighbors	Cos	PCA
Winogender (Rudinger et al., 2018)	✓	✓⃝	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓
Winobias (Zhao et al., 2018a)	✓	✓⃝	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓
Gap (Webster et al., 2018)			✓⃝	✓(aug)										
Crow-S (Nangia et al., 2020a)					✓⃝		✓		✓	✓	✓	✓	✓	✓
StereoSet (Nadeem et al., 2021)					✓⃝		✓		✓	✓	✓	✓	✓	✓
Bias in Bios (De-Arteaga et al., 2019)	✓⃝	✓	✓	✓(aug)	✓(aug)	✓(aug)	✓		✓	✓	✓	✓	✓	✓
EEC (Kiritchenko and Mohammad, 2018)						✓⃝	✓		✓	✓	✓	✓		
STS-B for genders (Beutel et al., 2020)					✓	✓⃝	✓	✓	✓	✓	✓	✓	✓	✓
Dev et al. (2020a) (NLI)				✓	✓	✓	✓		✓	✓	✓	✓	✓	✓
PTB, WikiText, CNN/DailyMail (Bordia and Bowman, 2019)							✓⃝		✓					
BOLD (Dhamala et al., 2021)							✓⃝							
Templates from May et al. (2019a)							✓	✓⃝	✓	✓	✓	✓	✓	✓
Templates from Kurita et al. (2019)							✓⃝	✓⃝	✓	✓	✓	✓	✓	✓
DisCo templates (Beutel et al., 2020)					✓		✓⃝		✓	✓	✓	✓	✓	✓
BEC-Pro templates (Bartl et al., 2020)					✓		✓⃝		✓	✓	✓	✓	✓	✓
English-German news corpus (Basta et al., 2021)							✓		✓	✓⃝	✓⃝	✓⃝	✓⃝	✓⃝
Reddit (Guo and Caliskan 2021, Voigt et al. 2018)							✓		✓⃝	✓	✓	✓	✓	✓
MAP (Cao and Daumé III, 2021)			✓⃝	✓					✓	✓	✓	✓	✓	✓
GICoref (Cao and Daumé III, 2021)			✓⃝						✓	✓	✓	✓	✓	✓

Table 1: Combinations of gender bias datasets and metrics in the literature. ✓ marks a feasible combination of a metric and a dataset. ✓⃝ marks the original metrics used on the dataset, and ✓(aug) marks metrics that can be measured after augmenting the dataset such that every example is matched with a counterexample of another gender. Extrinsic performance metrics depend on gold labels while extrinsic prediction metrics do not. A full description of the metrics is given in Appendix A.

sured (many ✓'s in the same row). Some datasets, such as Bias in Bios (De-Arteaga et al., 2019), have numerous metrics compatible, while others have fewer, but still multiple, compatible metrics. Bias in Bios has many compatible metrics since it has information that can be used to calculate them: in addition to gold labels, it also has gender labels and clear stereotype definitions derived from the labels which are professions. Text corpora and template data, which do not address a specific task (bottom seven rows), are mostly compatible with intrinsic metrics. The compatibility of intrinsic metrics with many datasets may explain why papers report intrinsic metrics more often (§5). Additionally, Table 1 indicates that not many datasets can be used to measure extrinsic metrics, particularly extrinsic performance metrics that require gold labels. On the other hand, measuring LM prediction on target words, which we consider as extrinsic, can be done on many datasets. This is useful for analyzing bias when dealing with LMs. It can be done by computing bias metrics from the LM output predictions, such as the mean probability gap when predicting the word "he" versus "she" in specific contexts. Also, some templates are valid for measuring extrinsic prediction metrics, especially stereotype-related metrics, as they were developed with explicit stereotypes in mind (such as profession-related stereotypes).

Based on Table 1, it is clear that there are many possible ways to measure gender bias in the literature, but they all fall under the vague category of "gender bias". Each of the possible combinations gives a different definition, or interpretation, for gender bias. The large number of different metrics makes it difficult or even impossible to compare different studies, including proposed gender bias mitigation methods. This raises questions about the validity of results derived from specific combinations of measurements. In the next two sections, we demonstrate how the choice of datasets and metrics can affect the bias measurement.

4 Effect of Dataset on Measured Results

The choice of data to measure bias has an impact on the calculated bias. Many researchers used sentence templates that are "semantically bleached" (e.g., "This is <word>.", "<person> studied <profession> at college.") to adjust metrics developed for static word embeddings to contextualized representations (May et al., 2019b; Kurita et al., 2019; Webster et al., 2020; Bartl et al., 2020). Delobelle et al. (2021) found that the choice of templates significantly affected the results, with little correlation between different templates. Additionally, May et al. (2019b) reported that templates are not as semantically bleached as expected.

Another common feature of bias metrics is the use of hand-curated word lexicons by almost every bias metric in the literature. Antoniak and Mimno (2021) reported that the lexicon choice can greatly affect bias measurement, leading to differing conclusions between different lexicons.

Metric	Testing balancing		
	Original	Oversampled	Subsampled
TPR (p)	0.78	0.75	0.75
TPR (s)	2.35	2.41	2.38
FPR (p)	0.61	0.59	0.57
FPR (s)	0.08	0.08	0.08
Precision (p)	0.71	-0.80	-0.89*
Precision (s)	1.87	2.59*	3.69*
Separation (s)	2.27	0.23*	0.35*
Sufficiency (s)	1.94	0.74*	9.15*
Independence (s)	0.14	0.01*	0.01*

Table 2: Metrics measured on Bias in Bios, separated to performance gap metrics (above the line) and statistical fairness metrics (below the line). Metrics are measured on the original test split, and on a subsampled and oversampled version of it. * marks statistically significant difference in a metric compared to the baseline (Original), using Pitman's permutation test ($p < 0.05$).

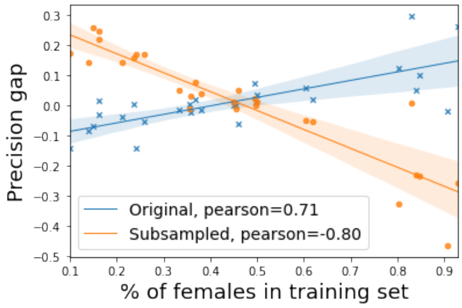

Figure 2: Percentage of females in the training set versus the resulting precision gap, per each profession. The trend is opposite on different test sets.

4.1 Case study: balancing the test data

Another important variable in gender bias evaluation, often overlooked in the literature, is the composition of the test dataset. Here, we demonstrate this by comparing metrics on different test sets, which come from the same dataset but have a different balance of examples. Bias in Bios (De-Arteaga et al., 2019) involves predicting an occupation from a biography text. These occupations are not balanced across genders, so for example over 90% of the nurses in the dataset identify as females.

Our case study extends the experiments done by Orgad et al. (2022). In their work, they tested a RoBERTa-based (Liu et al., 2019) classifier finetuned on Bias in Bios. The model was trained and evaluated on a training/test split of the dataset using numerous extrinsic bias metrics. Here we train the same model on the same training set, but evaluate it on three types of test sets: the original test set alongside two balanced versions of it, which have equal numbers of females and males in every profession, by either subsampling or oversampling. [2] We follow Orgad et al. and report nine different metrics on this task, measuring either some notion of performance gap or a statistical metric from the fairness literature. For details on the metrics measured in this experiments, see Appendix C.1.

As the results in Table 2 show, although many of the gap metrics (top block) are unaffected by the balancing of the test dataset, the absolute sum of precision gaps is significantly different when tested on different datasets. Moreover, the Pearson correlation for precision is negative when tested on a balanced dataset (either subsampled or oversampled), which may lead to different conclusions. The Pearson correlation is computed between the performance gaps per label (profession), and the percentage of females in the training set for that label, without balancing (the original distribution can be found in Appendix C.3). Strong correlations (positive or negative) indicate more bias was learned from the training set, and the direction of bias is indicated by the sign. This correlation is illustrated in Figure 2 for the original (imbalanced) test set and a balanced (subsampled) test set. Appendix C.4 discusses why this occurs.

The statistical fairness metrics (bottom block in Table 2) show a significant difference in the measured bias across different test set balancing. Oversampling shows less bias than when measured on the original test set, while subsampling yields mixed results – it decreases one metric while increasing another.

What is the “correct” test set? Since metrics are defined over the entire dataset, they are sensitive to its composition. For measuring bias in a model, the dataset used should be as unbiased as possible, thus balanced datasets are preferable.

If we were only concerned with measuring one of the reduced metrics on a non-balanced test set, we could misrepresent the fairness of the model. Indeed, it is common practice to measure only a small portion of metrics out of all those that can be measured—as we show in section 5—which makes us vulnerable to misinterpretations.

[2] Subsampling is the process of removing examples from the dataset such that the resulting dataset contains the same number of male and female examples for each label. Oversampling achieves this by repeating examples.